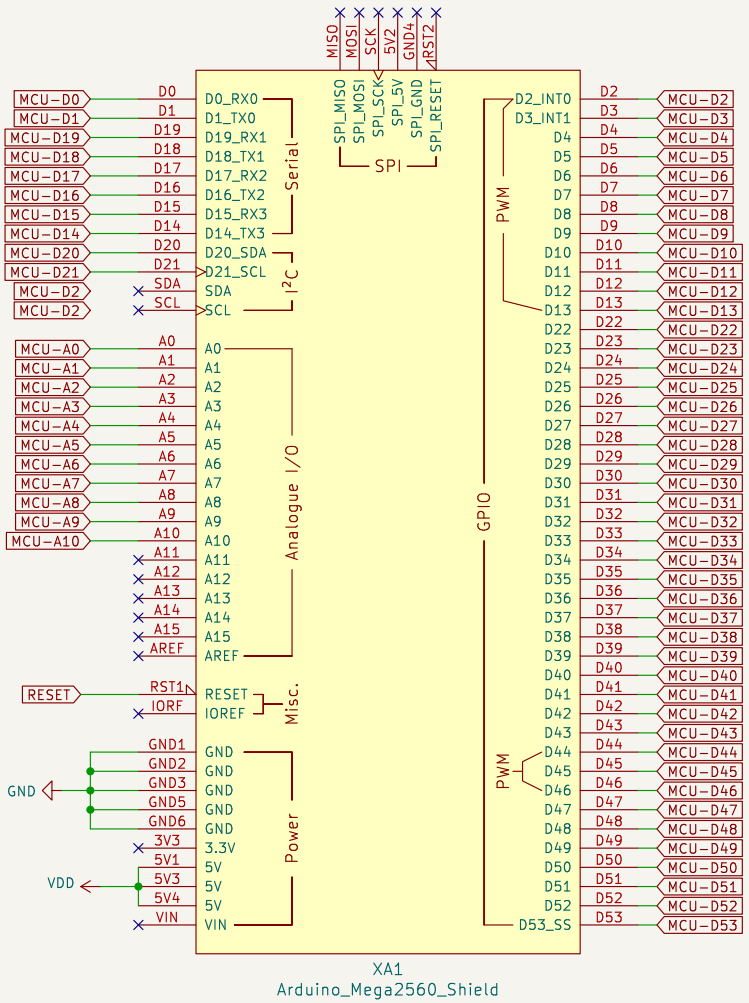


Fuel & Ignition
File: Fuel_Ignition.kicad_sch

Inputs
File: inputs.kicad_sch

Power
File: power.kicad_sch

Connector
File: connector.kicad_sch

Outputs
File: outputs.kicad_sch

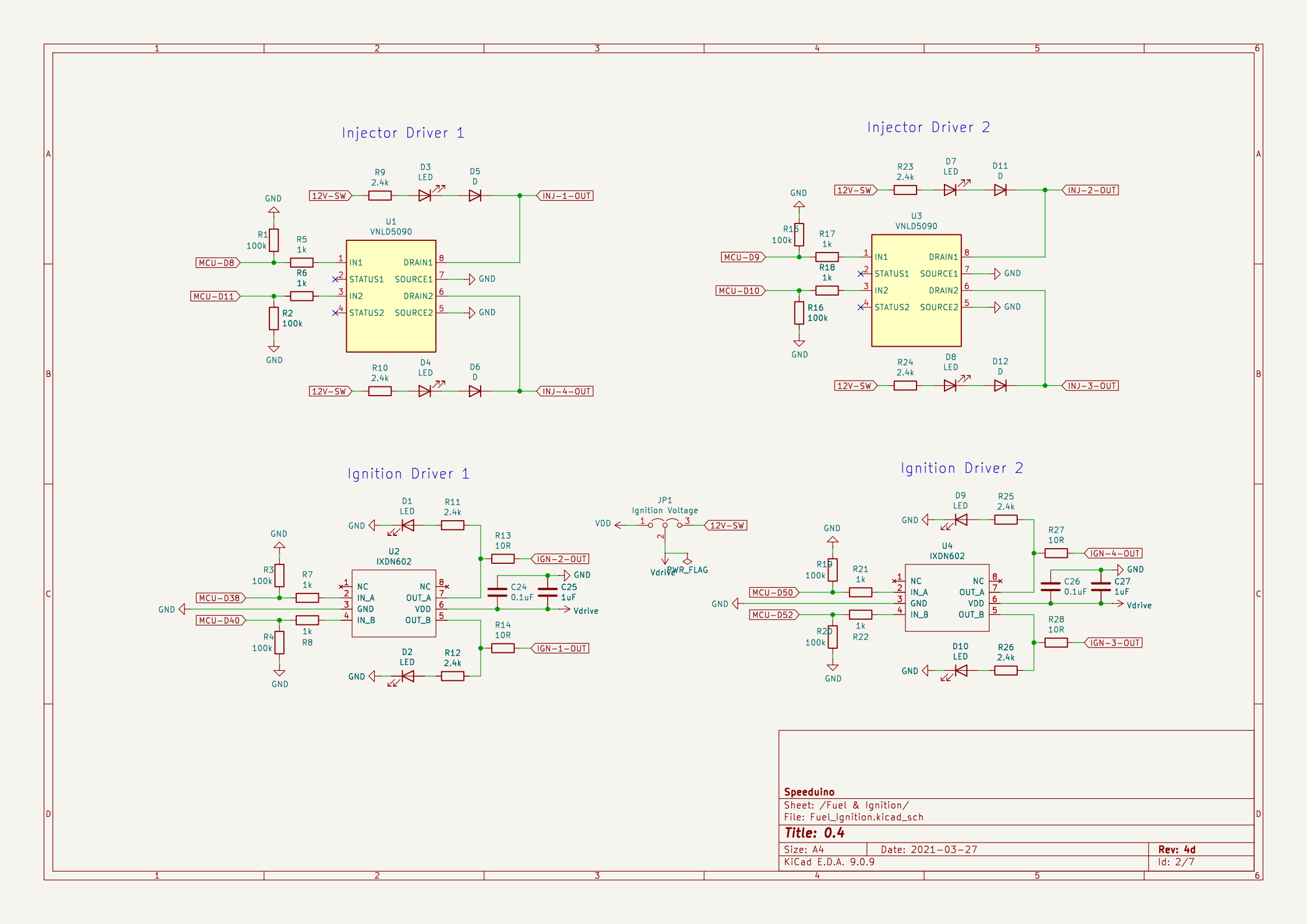
Proto Area
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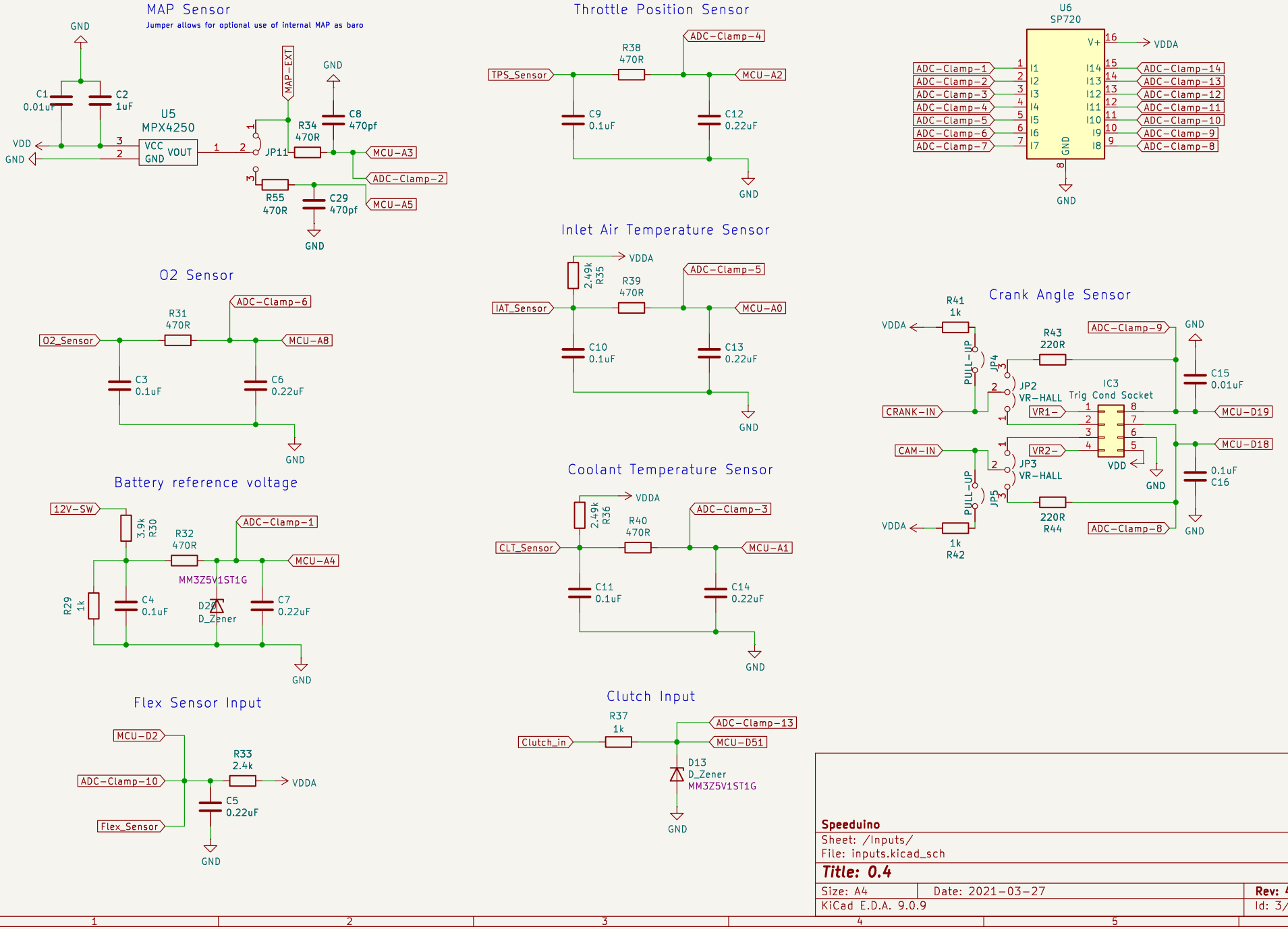
Speeduino

Sheet: /
File: v0.4.4d.kicad_sch

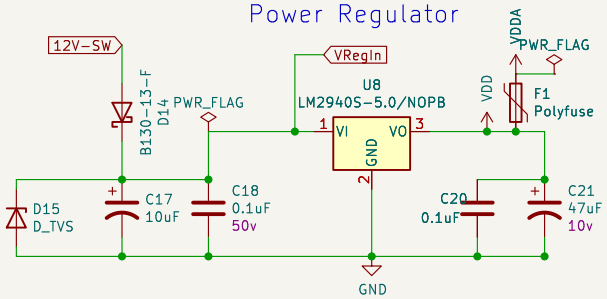
Title: 0.4

Size: A4	Date: 2021-03-27	**Rev: 4d**
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Injector Driver 1
R9
2.4k
D3
LED
D5
D
12V-SW
INJ-1-OUT
GND
R1
100k
R5
1k
U1
VNLD5090
MCU-D8
R6
1k
IN1
DRAIN1
STATUS1
SOURCE1
GND
MCU-D11
IN2
DRAIN2
R2
100k
STATUS2
SOURCE2
GND
GND
R10
2.4k
D4
LED
D6
D
12V-SW
INJ-4-OUT
Injector Driver 2
R23
2.4k
D7
LED
D11
D
12V-SW
INJ-2-OUT
GND
R15
100k
R17
1k
U3
VNLD5090
MCU-D9
R18
1k
IN1
DRAIN1
STATUS1
SOURCE1
GND
MCU-D10
IN2
DRAIN2
R16
100k
STATUS2
SOURCE2
GND
GND
R24
2.4k
D8
LED
D12
D
12V-SW
INJ-3-OUT
Ignition Driver 1
D1
LED
R11
2.4k
GND
R13
10R
IGN-2-OUT
U2
IXDN602
R3
100k
R7
1k
GND
NC
NC
MCU-D38
IN_A
OUT_A
C24
0.1uF
C25
1uF
GND
GND
GND
VDD
Vdrive
MCU-D40
IN_B
OUT_B
R14
10R
R4
100k
1k
R8
IGN-1-OUT
GND
D2
LED
R12
2.4k
GND
JP1
Ignition Voltage
VDD
12V-SW
Vdrive
PWR_FLAG
Ignition Driver 2
D9
LED
R25
2.4k
GND
R27
10R
IGN-4-OUT
U4
IXDN602
GND
R19
100k
R21
1k
NC
NC
GND
C26
0.1uF
C27
1uF
MCU-D50
IN_A
OUT_A
GND
Vdrive
MCU-D52
GND
VDD
R28
10R
R20
100k
1k
R22
IN_B
OUT_B
IGN-3-OUT
GND
D10
LED
R26
2.4k
GND
Speeduino
Sheet: /Fuel & Ignition/
File: Fuel_Ignition.kicad_sch
Title: 0.4
Size: A4
Date: 2021-03-27
Rev: 4d
KiCad E.D.A. 9.0.9
Id: 2/7

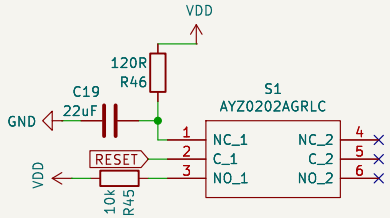

MAP Sensor
Jumper allows for optional use of internal MAP as baro
GND
C1
0.01uF
C2
1uF
U5
MPX4250
VDD
GND
VCC
GND
VOUT
MAP-EXT
JP11
R34
470R
C8
470pf
MCU-A3
ADC-Clamp-2
R55
470R
C29
470pf
MCU-A5
Throttle Position Sensor
TPS_Sensor
R38
470R
ADC-Clamp-4
MCU-A2
C9
0.1uF
C12
0.22uF
U6
SP720
V+
VDDA
ADC-Clamp-1
ADC-Clamp-2
ADC-Clamp-3
ADC-Clamp-4
ADC-Clamp-5
ADC-Clamp-6
ADC-Clamp-7
ADC-Clamp-14
ADC-Clamp-13
ADC-Clamp-12
ADC-Clamp-11
ADC-Clamp-10
ADC-Clamp-9
ADC-Clamp-8
O2 Sensor
O2_Sensor
R31
470R
ADC-Clamp-6
MCU-A8
C3
0.1uF
C6
0.22uF
Inlet Air Temperature Sensor
IAT_Sensor
2.49k
R35
R39
470R
ADC-Clamp-5
MCU-A0
C10
0.1uF
C13
0.22uF
Crank Angle Sensor
R41
1k
R43
220R
ADC-Clamp-9
C15
0.01uF
PULL-UP
JP4
JP2
VR-HALL
IC3
Trig Cond Socket
CRANK-IN
VR1-
MCU-D19
CAM-IN
VR2-
MCU-D18
JP3
VR-HALL
JP5
C16
0.1uF
220R
R44
ADC-Clamp-8
1k
R42
Battery reference voltage
12V-SW
3.9k
R30
R32
470R
ADC-Clamp-1
MCU-A4
MM3Z5V1ST1G
R29
1k
C4
0.1uF
D2
D_Zener
C7
0.22uF
Coolant Temperature Sensor
CLT_Sensor
2.49k
R36
R40
470R
ADC-Clamp-3
MCU-A1
C11
0.1uF
C14
0.22uF
Flex Sensor Input
MCU-D2
ADC-Clamp-10
Flex_Sensor
R33
2.4k
C5
0.22uF
Clutch Input
Clutch_in
R37
1k
ADC-Clamp-13
MCU-D51
D13
D_Zener
MM3Z5V1ST1G
Speeduino
Sheet: /Inputs/
File: inputs.kicad_sch
Title: 0.4
Size: A4
Date: 2021-03-27
Rev: 4d
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Power Regulator

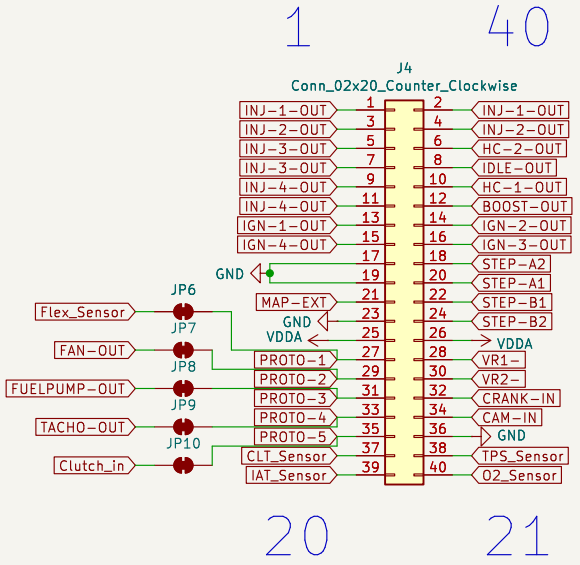


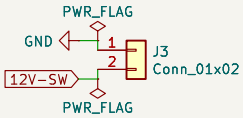
Reset Protection

Refer AVR040 Application Note



Speeduino		
Sheet: /Power/		
File: power.kicad_sch		
Title: 0.4		
Size: A4	Date: 2021-03-27	**Rev: 4d**
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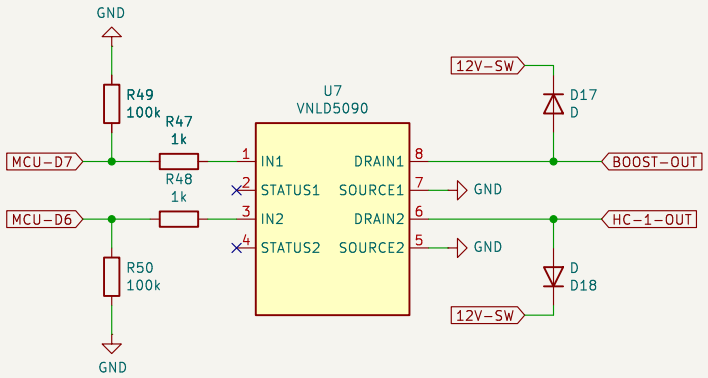
Speeduino

Sheet: /Connector/
File: connector.kicad_sch

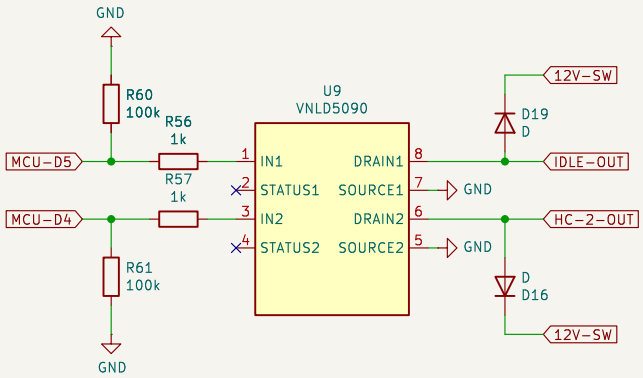
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Size: A4	Date: 2021-03-27	**Rev: 4d**
KiCad E.D.A. 9.0.9		Id: 5/7

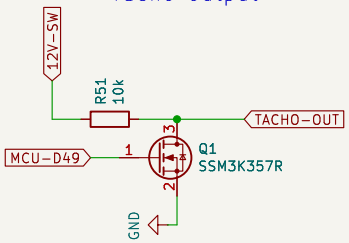
PWM Idle and Boost outputs



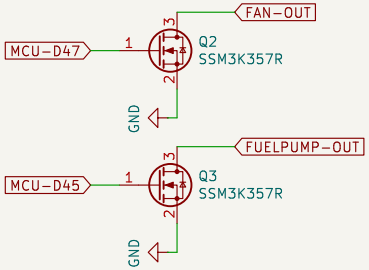
Spare High Current outputs



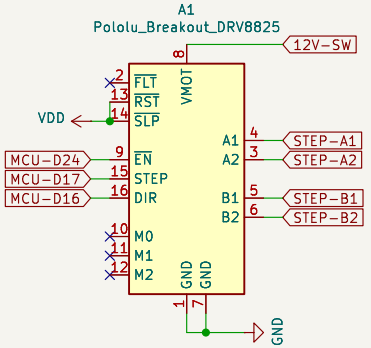
Tacho output



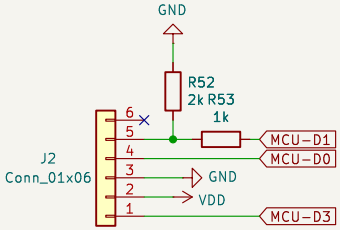
Fan and Fuel Pump outputs



Stepper Idle



HC-05 Bluetooth Connector

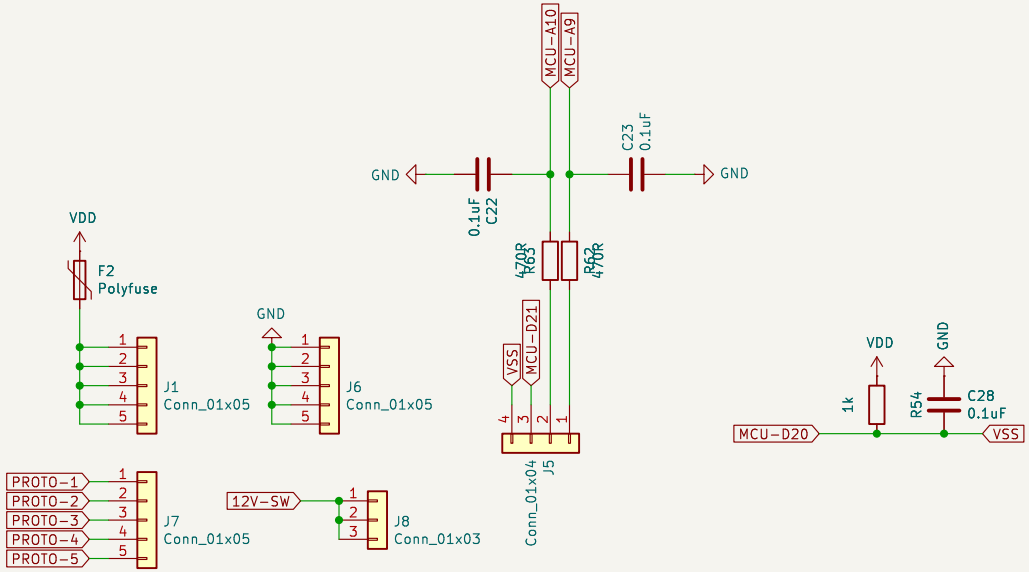


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Sheet: /Outputs/
File: outputs.kicad_sch

Title: 0.4

Size: A4	Date: 2021-03-27	**Rev: 4d**
KiCad E.D.A. 9.0.9		Id: 6/7



Speeduino

Sheet: /Proto Area/
File: proto.kicad_sch

Title: 0.4

Size: A4	Date: 2021-03-27	**Rev: 4d**
KiCad E.D.A. 9.0.9		Id: 7/7